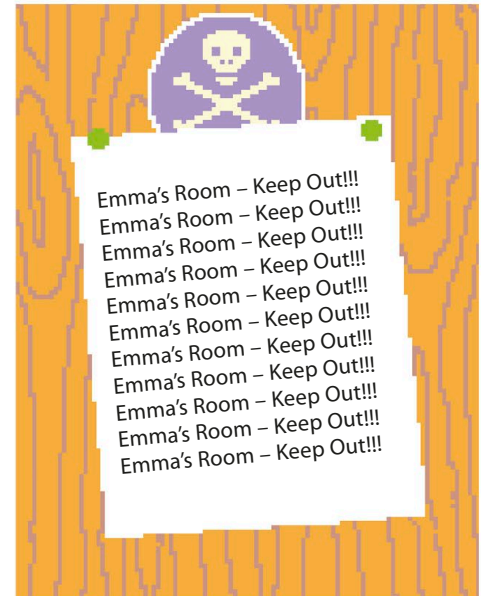
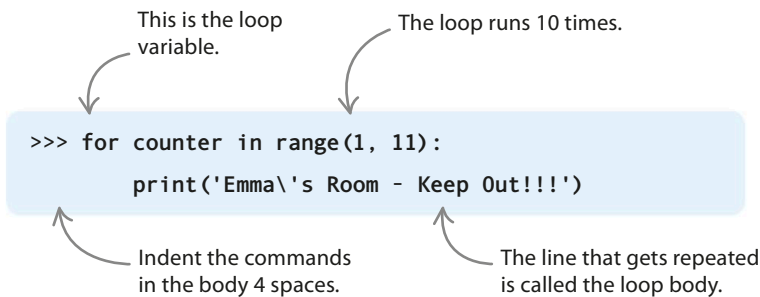


Loopy loops

Computers are great at doing boring tasks without complaining. Programmers aren't, but they are good at getting computers to do repetitive work for them – by using loops. A loop runs the same block of code over and over again. There are several different types of loop.

For loops

When you know how many times you want to run a block of code, you can use a **for** loop. In this example, Emma has written a program to make a sign for her door. It prints “Emma’s Room – Keep Out!!!” ten times. Try out her code for yourself in the shell. (After typing the code and hitting enter/return, press backspace to remove the indent and then hit enter/return again.)



▽ Loop variable

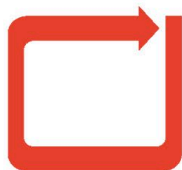
The loop variable keeps track of how many times we've gone round the loop so far. The first time round it's equal to the first number in the list specified by **range(1, 11)**. The second time round it's equal to the second number in the list, and so on. When we've used all the numbers in the list, we stop looping.

First loop



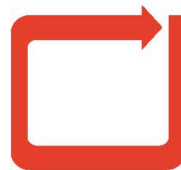
Loop variable = 1

Second loop



Loop variable = 2

Third loop



Loop variable = 3



EXPERT TIPS

Range

In Python code, the word “range” followed by two numbers within brackets stands for “all the numbers from the first number to one less than the second number”. So **range(1, 4)** means the numbers 1, 2, and 3 – but not 4. In Emma’s “Keep Out” program, **range(1, 11)** is the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10.